

Reminders:

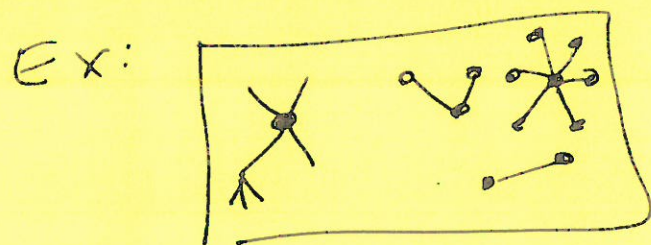
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Tree (in graph theory): a graph that has no cycles (aka $\nexists$ a nontrivial path that starts/ends at the same place)

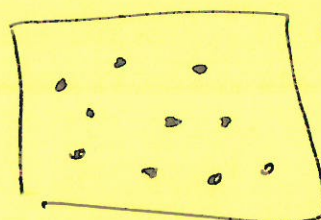
ex: binary search tree: root + 2 children per node (unless a leaf node)

parent/child relationship

Forest: a set of trees



Another
EX:



Minimum Spanning Tree:

Given a graph $G = (V, E, w)$
a tree $T = (V_T, E_T)$ such that a MST for G is

(i) $V_T = V$ (spans all the vertices)

(ii) of all spanning trees, this has minimum cost

$$c(T) = \sum_{e \in E_T} w(e)$$

Connected components: can partition V into CC's
• v, w are in the same connected comp if $\exists$ path from v to w .

①

Question 2:

What is "the issue"?



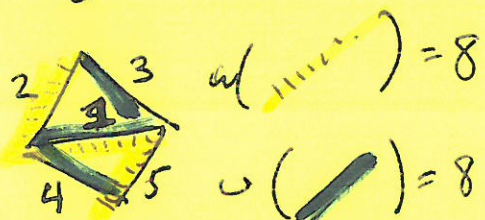
or



} How to pick among "the MSTs" so we're always finding the same one.
We can't

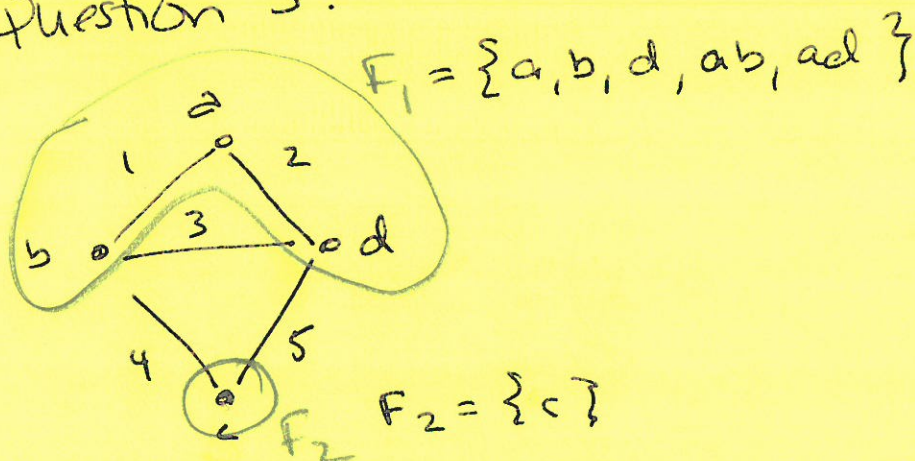
There is no canonical MST.

So, the assumption that edge weights are unique will work.



$$w\left(\begin{array}{c} 2 \\ \swarrow \quad \searrow \\ 1 \\ \swarrow \quad \searrow \\ 4 \end{array}\right) = 7$$

Question 3:

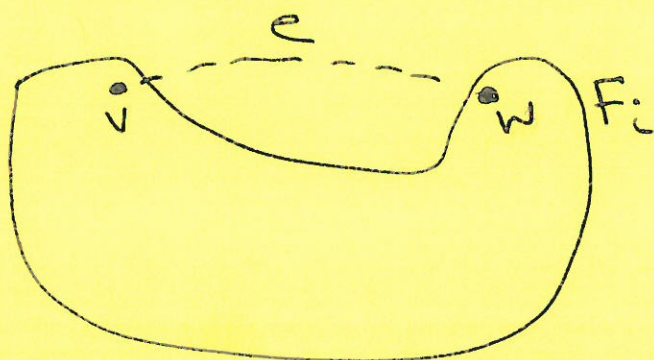


$$\begin{aligned} E \setminus F &= \{ab, ad, bd, bc, cd\} \setminus \{ab, ad\} \\ \uparrow \quad \uparrow \\ \text{edge set} \quad \text{graph} &= \{bd, bc, cd\} \\ &\quad \uparrow \quad \underbrace{\hspace{2cm}} \\ &\quad \checkmark \quad \text{different conn. components.} \\ e = \{bd\} \text{ meets criteria} &\rightarrow \text{but not in MST.} \end{aligned}$$

Question 3

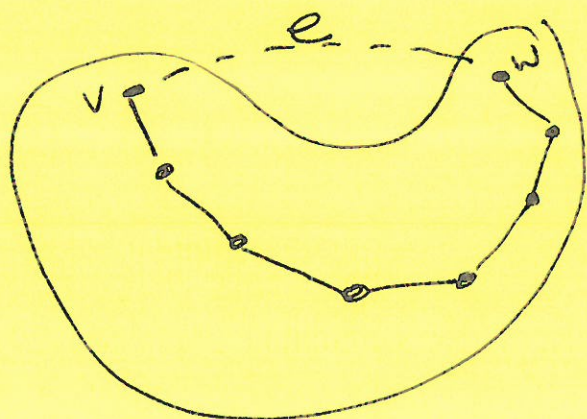
Why are useless edges useless?

Consider F_i the connected comp containing the endpoints of $e = (v, w)$



Because v, w are in same conn comp

$\Rightarrow \exists$ a path from v to w in F_i .



(length > 1 b/c otherwise, we'd have a double edge)

$F_i \cup \{e\}$ contains a loop $\Rightarrow$ not a subgraph of MST

know all edges of F_i are in MST
 $\Rightarrow e \notin \text{MST}$.

CSCI 432 Handout 14: Minimum Spanning Trees

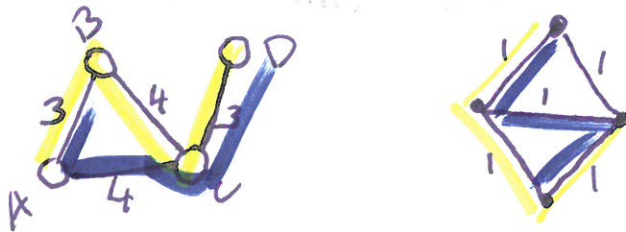
Name: CLASS

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Examples

1. Draw an example of a weighted graph with two distinct minimum spanning trees.

Answer



2. What assumptions will remove this issue?

Answer

on the input graph

well, it's not an "issue" per say, but would be nice if the MST were unique.

OPT-① Consider Graphs w/ no cycles $\Rightarrow$ trees
MST of a tree is itself.

(a bit too boring algorithmically)

OPT-② Graphs are connected (otw, wouldn't have a spanning tree)
+ undirected
+ all edge weights are distinct.

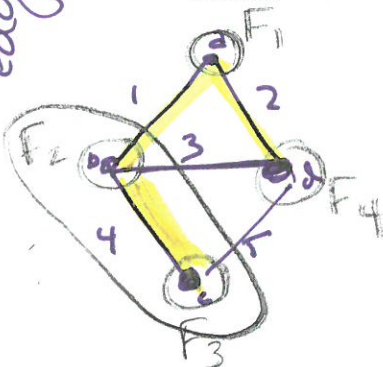
3. For the next two questions, suppose we have a weighted graph $G = (V, E, \omega)$ meeting the assumptions listed above whose minimum spanning tree is T . Let F be a subgraph of T .

If $e \in E \setminus F$ and both endpoints of e are in the same component of F , is $e \in T$? Why or why not?

Answer

$\rightarrow$ a forest!
 $\rightarrow$ NO!
 $\rightarrow$ ie, same maximal subtree

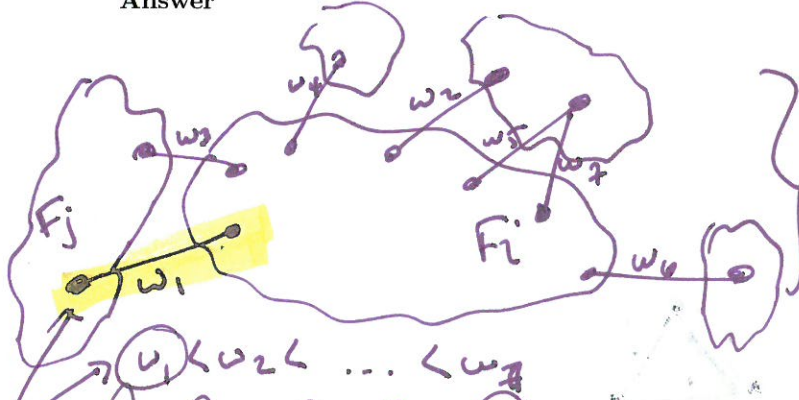
$$F1 = F_1 \cup F_2 \cup \{F_2 \cup F_3 \cup [bc]\}$$



$F = F_1 \cup F_2 \cup F_3 \cup F_4$ } each F_i is a tree w/ 1 vertex
 $\therefore F$ has no edges

4. Let $E' \in E \setminus F$ be the set of edges such that their endpoints are in different components of F . Let e be the minimum-weight edge in E' . Is $e \in T$? Why or why not?

Answer



5. What assumptions will remove this issue?

Answer

Safe edge for the 7th component

Borůvka's Algorithm

In the textbook, Dr. Erickson describes a generic MST algorithm: We start with F as the set of vertices (a spanning forest, if you will), and add edges to R until we find a MST.

In Borůvka's algorithm, all safe edges are added at each step.

1. Walk through an example, marking all safe edges and all useless edges at each step. (The graph you start with should have at least eight vertices and 10 edges, and should satisfy the assumptions we listed on the first page).

Answer

2. What is an observation you have about the useless edges?

Answer

3. What is the loop invariant for this algorithm?

Answer

Jarnik's / Prim's Algorithm

In Jarnik's algorithm, we have a subtree T' of the MST T . We obtain this by starting at an arbitrary vertex and adding the minimum safe edge for T' at each step.

1. Walk through an example. (The graph you start with should have at least eight vertices and 10 edges, and should satisfy the assumptions we listed on the first page).

Answer

2. What is the loop invariant for this algorithm?

Answer

3. What are some pros/cons of using Jarník's versus Borůvka's algorithms?

Answer

Kruskal's Algorithm

In Kruskal's algorithm, we again have a spanning forest F at each stage, initialized to each vertex is its own tree. At each step, we pick the minimum weight edge and add it to F , as long as it is not useless.

1. Walk through an example. (The graph you start with should have at least eight vertices and 10 edges,

and should satisfy the assumptions we listed on the first page).

Answer